5.5 For each of the following two time-series regression models, and assuming MR1–MR6 hold, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T\to\infty}\text{var}(b_2|\mathbf{x}) = 0$.

a. $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \ldots, T$. Note that $\mathbf{x} = (1, 2, \ldots, T)$, $\sum_{t=1}^{T}(t-\bar{t})^2 = \sum_{t=1}^{T} t^2 - \left(\sum_{t=1}^{T} t\right)^2 / T$, $\sum_{t=1}^{T} t = T(T+1)/2$ and $\sum_{t=1}^{T} t^2 = T(T+1)(2T+1)/6$.

b. $y_t = \beta_1 + \beta_2(0.5)^t + e_t, t = 1, 2, \ldots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \ldots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \cdots + r^n = \frac{r(1-r^n)}{1-r}$$

c. Provide an intuitive explanation for these results.

a.

$$\sum_{t=1}^{T}(t-\bar{t})^2 = \sum t^2 - \frac{(\sum t)^2}{T}$$

$$= \frac{T(T+1)(2T+1)}{6} - \frac{\left[\frac{T(T+1)}{2}\right]^2}{T}$$

$$= \frac{T(T+1)(2T+1)}{6} - \frac{T(T+1)^2}{4}$$

$$= \frac{T(T+1)}{2}\left[\frac{2T+1}{3} - \frac{T+1}{2}\right]$$

$$= \frac{T(T+1)}{2}\left[\frac{4T+2-3T-3}{6}\right]$$

$$= \frac{T(T+1)}{2}\left[\frac{T-1}{6}\right]$$

$$= \frac{T(T+1)(T-1)}{12} = \frac{T(T^2-1)}{12}$$

$$Var(b_2|x) = \frac{\sigma^2}{\sum_{t=1}^{T}(x_t-\bar{x})^2}$$

$$= \frac{\sigma^2}{\frac{T(T^2-1)}{12}} = \frac{12\sigma^2}{T(T^2-1)}$$

$$\therefore \lim_{T\to\infty} var(b_2|x) = \lim_{T\to\infty}\frac{12\sigma^2}{T^3-T} = 0$$

b. $x_t = 0.5^t$

$$\sum(x_t-\bar{x})^2 = \sum x_t^2 - \frac{(\sum x_t)^2}{T}$$

① $$\sum x_t^2 = \sum(0.5^t)^2 = \sum_{1}^{T} 0.25^t$$

$$= \frac{0.25(1-0.25^T)}{1-0.25} = \frac{0.25}{0.75}(1-0.25^T)$$

$$= \frac{1}{3}(1-0.25^T)$$

② $$\sum^{T} x_t = \sum 0.5^t = \frac{0.5(1-0.5^T)}{1-0.5}$$

$$= \frac{0.5}{0.5}(1-0.5^T) = 1-0.5^T$$

$$\Rightarrow \sum(x_t-\bar{x})^2 = \frac{1}{3}(1-0.25^T) - \frac{(1-0.5^T)^2}{T}$$

$$Var(b_2|x) = \frac{\sigma^2}{\underbrace{\frac{1}{3}(1-0.25^T)}_{0} - \cancel{\frac{(1-0.5^T)^2}{T}}\ 0}$$

$$T\to\infty,\ 0.25^T\to 0,\ 0.5^T\to 0,\ \frac{1}{T}\to 0$$

$$\lim_{T\to\infty} var(b_2|x) = \frac{\sigma^2}{\frac{1}{3}} = 3\sigma^2 \neq 0.$$

c. $x_t = t$ represents a linear time trend. As T grows, the values keep expanding, causing the total variation of x to approach infinity. This means every new observation brings substantial new information, effectively driving the estimation variance to 0. Hence, it's consistent.

$x_t = 0.5^t$ decays toward 0 extremely rapidly. After the first observation, subsequent values of x_t are pratically zero. Therefore, adding more data points ($T\to\infty$) adds virtually no new variation or information about how x affects y. Since the total variation in x hits an asymptote (it caps at $\frac{1}{3}$), the uncertainty in our estimate never completely vanishes (remaining stuck at $3\sigma^2$). Hence, it's inconsistent.

5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$K = 3$

$N - K = 60$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \qquad \widehat{\operatorname{cov}}(b_1, b_2, b_3) = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix} \end{array}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

a. $\beta_2 = 0$
b. $\beta_1 + 2\beta_2 = 5$
c. $\beta_1 - \beta_2 + \beta_3 = 4$

a. $t\text{-statistic} = \dfrac{b_2 - \beta_2}{se(b_2)} = \dfrac{3}{2} = 1.5$

$t_{60,\,0.975} = 2$

$\because 1.5 < 2$

$\therefore$ Do not reject H_0.

b.

$t\text{-statistic} = \dfrac{b_1 + 2b_2 - 5}{se(b_1 + 2b_2)} = \dfrac{3}{\sqrt{11}} \approx 0.9045$

$Var(b_1 + 2b_2) = Var(b_1) + 4\,Var(b_2) + 4\,cov(b_1, b_2)$

$= 3 + \underbrace{4(4)}_{16} + \underbrace{4(-2)}_{-8}$

$= 11$

$\because 0.9045 < 2 \quad \therefore$ Do not reject H_0.

c.

$t\text{-statistic} = \dfrac{(b_1 - b_2 + b_3) - 4}{se(b_1 - b_2 + b_3)}$

$= \dfrac{(2-3-1)-4}{\sqrt{16}} = \dfrac{-6}{4} = -1.5$

$Var(b_1 - b_2 + b_3) = Var(b_1) + Var(b_2) + Var(b_3)$

$-2cov(b_1, b_2) - 2cov(b_2, b_3) + 2cov(b_1, b_3)$

$= 3 + 4 + 3 - 2(-2) - 2(0) + 2(1)$

$= 16$

$\because |-1.5| < 2$

$\therefore$ Do not reject H_0 #

5.20 A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i$, $i = 1, 2, \ldots, N$ is

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x. Given that assumptions MR1–MR6 hold:

a. Show that $\hat{\beta}_{2,mean}$ is unbiased.

b. Derive an expression for $\operatorname{var}(\hat{\beta}_{2,mean}|\mathbf{x})$.

c. Write down an expression for $\operatorname{var}(\hat{\beta}_{2,mean})$.

d. Under what conditions will $\hat{\beta}_{2,mean}$ be a consistent estimator for β_2?

e. Randomly generate observations on x from a uniform distribution on the interval (0,10) for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:

   i. $\operatorname{var}(b_2|\mathbf{x})$ and $\operatorname{var}(\hat{\beta}_{2,mean}|\mathbf{x})$ where b_2 is the OLS estimator.

   ii. Estimates for $E\left[(s_x^2)^{-1}\right]$ and $E\left[4/(\bar{x}_2 - \bar{x}_1)^2\right]$ where s_x^2 is the sample standard deviation for x using N as a divisor.

f. Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2,mean}$ is consistent?

g. Show that $E\left[(s_x^2)^{-1}\right] \xrightarrow{p} 0.12$ and $E\left[4/(\bar{x}_2 - \bar{x}_1)^2\right] \xrightarrow{p} 0.16$. [*Hint*: The variance of a uniform random variable defined on the interval (a, b) is $(b - a)^2/12$.]

h. Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

a.

$$\bar{y}_1 = \beta_1 + \beta_2 \bar{x}_1 + e_1$$

$$\bar{y}_2 = \beta_1 + \beta_2 \bar{x}_2 + e_2$$

$$\hat{\beta}_{2,mean} = \frac{(\beta_1 + \beta_2 \bar{x}_2 + e_2) - (\beta_1 + \beta_2 \bar{x}_1 + e_1)}{\bar{x}_2 - \bar{x}_1}$$

$$= \frac{\beta_2(\bar{x}_2 - \bar{x}_1) + (e_2 - e_1)}{\bar{x}_2 - \bar{x}_1} = \beta_2 + \frac{e_2 - e_1}{\bar{x}_2 - \bar{x}_1}$$

$$E(\hat{\beta}_{2,mean}|X) = \beta_2 + \frac{\overset{0}{E(e_2|X)} - \overset{0}{E(e_1|X)}}{\bar{x}_2 - \bar{x}_1}$$

According to MR2, $E(e_i|X) = 0$.

$$\therefore E(\hat{\beta}_{2,mean}|X) = \beta_2 \quad *$$

b. $$\hat{\beta}_2 - \beta_2 = \frac{e_2 - e_1}{\bar{x}_2 - \bar{x}_1}$$

$$\operatorname{var}(\hat{\beta}_2|X) = \operatorname{var}\left(\frac{e_2 - e_1}{\bar{x}_2 - \bar{x}_1}\Big|X\right) = \frac{\operatorname{var}(e_2|X) + \operatorname{var}(e_1|X) - 2\operatorname{cov}(e_1, e_2|X)}{(\bar{x}_2 - \bar{x}_1)^2}$$

$$= \frac{\frac{2\sigma^2}{N} + \frac{2\sigma^2}{N}}{(\bar{x}_2 - \bar{x}_1)^2} = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \quad *$$

$\operatorname{var}(e_i) = \sigma^2$, $\operatorname{cov}(e_i, e_j) = 0$.

$$\operatorname{var}(e_1|X) = \operatorname{var}\left(\frac{1}{\frac{N}{2}}\sum_{i=1}^{N/2} e_i\right) = \frac{1}{(N/2)^2}\sum_1^{N/2} \operatorname{var}(e_i) = \frac{N/2 \cdot \sigma^2}{N^2/4} = \frac{2\sigma^2}{N}$$

同理 $\operatorname{var}(e_2|X) = \frac{2\sigma^2}{N}$

c.

$$\operatorname{var}(\hat{\beta}_{2,mean}) = E[\operatorname{var}(\hat{\beta}_{2,mean}|X)] + \operatorname{var}(E[\hat{\beta}_{2,mean}|X])$$

$E[\hat{\beta}_{2,mean}|X]$: constant, $\sigma^2 = 0$

$$\operatorname{var}(\hat{\beta}_{2,mean}) = E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right] = \frac{4\sigma^2}{N} E\left(\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right) *$$

d. $N \to \infty$ $\quad \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \to 0.$

This requires $N(\bar{x}_2 - \bar{x}_1)^2$ to go to infinity.

$(\bar{x}_2 - \bar{x}_1)$ need to coverage to a strictly positive constant.

f. 1. $\operatorname{var}(b_{2,OLS}) < \operatorname{var}(\hat{\beta}_{2,mean})$ always; aligns with Gauss-Markov Theorem, which states that OLS has smallest variance in all of the Linear Unbiased Estimator

2. As N increases, the variance of both gradually decreases and approaches toward 0, indicating that $\hat{\beta}_{2,mean}$ is consistent.

g. $E[(S_x^2)^{-1}] \xrightarrow{p} 0.12$

$E\left[\frac{4}{(\bar{X}_2-\bar{X}_1)^2}\right] \xrightarrow{p} 0.16$

pf. $U(0,10)$ $\sigma_x^2 = \frac{(10-0)^2}{12} = \frac{100}{12} \approx 8.333$

According to LLN,

$\frac{1}{S_x^2} \to \frac{1}{\sigma_x^2} = \frac{12}{100} = 0.12$

$\bar{X}_1 \xrightarrow{p} E[U(0,5)] = 2.5$

$\bar{X}_2 \xrightarrow{p} E[U(5,10)] = 7.5$

$(\bar{X}_2 - \bar{X}_1) \xrightarrow{p} 7.5 - 2.5 = 5$

$\frac{4}{(\bar{X}_2-\bar{X}_1)^2} \xrightarrow{p} \frac{4}{5^2} = \frac{4}{25} = 0.16$.

h.

If the observations are randomly assigned without ordering, both halves become random samples from the the population.

By LLN, both $\bar{X}_1$ and $\bar{X}_2$ coverage to the population mean $E[X]$.

$\bar{X}_1 \xrightarrow{p} E[X]$ $\bar{X}_2 \xrightarrow{p} E[X]$

Then $(\bar{X}_2 - \bar{X}_1) \xrightarrow{p} 0$, $\frac{4\sigma^2}{N(\bar{X}_2-\bar{X}_1)^2} \to \infty$

The estimator would be inconsistent, failing to capture the variation in X.